Brian Johns

DATA SCIENTIST · SPORTS ANALYTICS · 3× OLYMPIAN

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CREDENTIALS

- 3× Canadian Olympic Swimmer
- World Record Holder — 400m IM
- 25+ Years Elite Sport

CONTACT

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SKILLS

Languages
Python · SQL · R

ML / Modelling
Scikit-Learn · XGBoost
Keras · PyTorch · TensorFlow

Analysis
Pandas · NumPy · PySpark · SciPy

Visualisation
Tableau · Matplotlib
Seaborn · Plotly · Holistics

Cloud / Big Data
AWS · SageMaker · Spark · Hive

Statistics
Hypothesis Testing · A/B Testing
Bayesian Methods

Tools
Git · Jupyter · VS Code

EDUCATION

Diploma, Data Science
BrainStation · 2022

MSc Kinesiology
Coaching Science · UBC · 2011

BSc Human Kinetics
UBC · 2009

PROFILE

- Most data scientists understand performance data analytically. I've lived it as a 3× Canadian Olympian and World Record holder, applied it as a national-level coach, and built with it in a start-up environment. The rare combination of elite athletic intelligence, coaching science, and data skills has consistently translated into success in the water and in production.

EXPERIENCE

Head of Coaching Science
FORM Swim · Vancouver, BC — Oct 2022 – Present

- Validated **FORM Score** — a 1–100 composite swimming efficiency metric normalised for height and pool distance — and developed the content strategy supporting its launch across the app and marketing.
- Built the **Swimmer Type classification system** in Python/SQL — classifies swimmers into 8 types using FORM Score, DPS, and Stroke Rate to drive personalised HeadCoach recommendations.
- Engineered the **HeadCoach Workout engine** — Python script selecting from 200+ JSON templates using athlete goals, performance data, and Swimmer Type. Became a leading indicator of trial-to-subscription conversion.
- Designed **HeadCoach Plans** — adaptive multi-week training plans across three pathways: race preparation, health & fitness goals, and general swimming improvement; contributed to measurable gains in subscriber retention.
- Designed the logic and content framework for **HeadCoach 2.0** — an expanded personalised coaching system integrating Swimmer Type, training objectives, and real-time technique tracking to deliver bespoke in-goggle feedback and post-swim insights at scale.

Teacher's Assistant — Data Science
BrainStation · Vancouver, BC — Apr 2022 – Jul 2022

- Selected based on academic performance to mentor students in SQL, Tableau, Python, ML, and big data analysis.
- Taught supplementary sessions on advanced Python usage and ML model implementation.

Director of Swimming / Head Coach
Vancouver Pacific Swim Club · Vancouver, BC — Sep 2016 – Oct 2021

- Led athlete development and stakeholder management across coaches, parents, board members, and facilities to deliver programming and meet financial expectations.
- Maintained operational continuity and financial stability through the pandemic, preserving programming during community instability.
- Built performance tracking frameworks bridging coaching science and data analysis for long-term athlete development.

Assistant Coach
UBC Thunderbirds · Vancouver, BC — Sep 2011 – Aug 2016

- Coached varsity athletes to provincial, national, and international level. Named **Junior Coach of the Year, 2012**.
- Built data frameworks to identify talent, guide recruitment, and maximise athlete potential.

National Team Swimmer · 3× Olympian
Swimming Canada — 2000 – 2012

- Represented Canada at three Olympic Games (2000, 2004, 2008).
- In 2003, broke the **World Record** in the 400m Individual Medley.
- Developed deep analytical understanding of performance metrics, physiological modelling, and athletic optimisation.

INDEPENDENT PROJECTS

Women's Hockey xG Model
Python · API Scraping · XGBoost · Logistic Regression · Bayesian Feature Engineering · Sports Analytics

Scraped PWHL play-by-play data across all available seasons and merged with PWHPA data to build a Women's professional hockey xG model. Engineered Bayesian shooter/goalie quality priors, strength state features, and shot location polygons.

NHL Player Salary Prediction & Evaluation
Python · XGBoost · Random Forest · Lasso Regression · K-Means Clustering · SHAP

Predicted and evaluated NHL cap hits using basic and advanced statistics. Identified market inefficiencies — defensive metrics and advanced stats remain undervalued. Evaluated the importance of star player depth in team roster building.

Mavericks Tweet Sentiment Analysis
Python · NLP · SMOTE · Logistic Regression · Naive Bayes · Multi-class Classification

Combined Twitter data with Dallas Mavericks game results to perform multi-class sentiment analysis using NLP. Found that negative sentiment following a loss is stronger and more persistent than positive sentiment after a comparable win.